

LENAWEE COUNTY AP, MI, USA

WMO: 725404

Lat: 41.868N

Lon: 84.079W

Elev: 796

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	4.8	-7.8	3.7	0.4	-2.0	5.1	6.7	24.3	27.7	21.7	26.7	5.9	240	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	90.2	72.9	87.6	71.9	84.4	70.3	76.2	85.2	74.6	82.9	73.1	80.9	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	127.4	80.7	72.1	122.4	79.4	70.4	115.3	77.6	40.1	85.2	38.5	82.7	37.1	80.9	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	18.9	17.0	-7.4	93.5	6.9	3.3	-12.3	95.9	-16.4	97.9	-20.3	99.7	-25.3	102.1
			-7.8	78.5	6.5	1.7	-12.5	79.7	-16.3	80.7	-20.0	81.6	-24.7	82.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	24.4	26.3	36.5	48.7	59.8	69.1	72.6	70.8	64.0	52.1	40.3	29.7	
	DBStd	19.13	11.29	10.53	10.71	8.86	8.27	6.33	5.61	5.04	7.23	8.67	8.90	9.75	
	HDD50	3060	795	664	440	126	13	0	0	0	1	80	311	630	
	HDD65	6321	1259	1083	885	492	206	31	6	11	104	410	741	1093	
	CDD50	2929	0	1	20	88	316	572	700	643	422	145	20	2	
	CDD65	717	0	0	1	4	44	153	241	190	74	10	0	0	
	CDH74	6360	0	0	10	62	463	1333	2245	1484	669	94	0	0	
	CDH80	1983	0	0	1	7	123	435	800	396	202	19	0	0	
Wind	WSAvg	7.7	8.9	8.7	8.8	9.6	7.8	6.8	5.9	5.5	6.1	7.5	8.1	8.5	
Precipitation	PrecAvg	33.3	1.9	1.7	2.4	3.1	3.3	3.6	3.1	3.4	3.3	2.3	2.8	2.5	
	PrecMax	45.1	4.3	4.8	4.3	7.0	8.0	6.9	7.1	8.6	9.4	6.7	6.4	5.1	
	PrecMin	19.5	0.1	0.3	0.5	0.9	0.9	0.6	0.6	0.1	0.3	0.1	0.7	0.5	
	PrecStd	5.2	1.1	1.1	1.0	1.3	1.8	1.6	1.5	2.1	1.8	1.3	1.3	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	60.5	73.9	80.7	88.1	92.3	94.9	91.1	90.4	83.0	69.6	58.1	
		MCWB	52.6	52.6	63.7	64.4	68.9	72.7	74.4	74.0	71.1	69.9	58.4	54.9	
	2%	DB	49.7	51.7	64.3	74.8	83.2	88.5	90.5	87.1	85.7	75.8	63.0	52.6	
		MCWB	47.5	48.5	55.7	60.2	66.8	71.6	74.0	72.8	70.1	63.5	55.4	49.8	
	5%	DB	43.9	45.0	58.2	70.1	79.6	85.0	87.8	84.1	81.4	71.4	58.5	47.2	
		MCWB	40.8	41.4	51.7	56.9	65.9	69.6	72.4	71.2	67.5	62.0	52.5	44.5	
	10%	DB	38.7	39.3	53.0	64.4	74.8	81.6	84.3	81.6	77.4	66.7	54.3	42.7	
		MCWB	36.2	35.9	47.5	54.2	63.7	68.6	70.9	70.3	66.4	59.2	49.5	39.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	54.4	64.0	67.1	72.7	76.0	78.5	78.1	75.7	70.8	61.6	55.3
			MCDB	55.2	57.2	74.1	75.9	82.9	87.1	89.1	87.0	85.3	80.4	66.5	57.3
2%		WB	47.7	47.9	57.6	63.0	69.8	74.0	76.6	75.8	73.0	66.9	56.9	50.1	
		MCDB	49.1	51.0	62.9	70.4	79.6	84.1	86.2	82.7	80.7	72.0	61.4	52.3	
5%		WB	41.4	41.2	53.1	59.1	67.9	72.4	74.9	74.0	70.6	63.4	53.4	44.8	
		MCDB	43.9	44.3	57.1	67.8	76.2	81.9	83.8	80.3	77.5	68.5	57.1	46.9	
10%		WB	36.5	36.4	47.6	55.4	65.6	70.5	73.2	72.2	68.4	60.1	49.7	40.0	
		MCDB	38.6	39.2	52.6	63.6	72.8	78.6	81.0	78.7	74.8	66.1	54.2	42.3	
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.4	16.7	20.4	20.4	19.8	20.4	19.6	21.2	18.9	15.4	12.1	
		MCDBR	16.9	19.8	25.0	28.0	25.9	24.0	23.3	22.2	25.9	25.0	22.2	16.4	
		MCWBR	15.1	16.4	17.7	16.3	13.1	10.9	9.5	10.2	12.3	14.1	16.0	14.6	
	5% WB	MCDBR	16.5	19.4	23.4	24.7	21.9	20.2	19.6	18.1	20.9	21.1	19.3	15.6	
		MCWBR	15.3	16.8	18.1	16.6	12.9	10.7	9.6	9.8	11.0	14.0	15.9	14.6	
Clear-Sky Solar Irradiance	taub	0.297	0.335	0.357	0.401	0.427	0.448	0.440	0.430	0.383	0.353	0.318	0.308		
	taud	2.396	2.270	2.322	2.218	2.194	2.177	2.216	2.232	2.347	2.420	2.473	2.396		
	Ebn at Noon	270	273	281	275	270	264	265	263	269	265	260	254		
	Edh at Noon	25	34	36	43	45	46	44	42	35	29	23	23		
All-Sky Solar Radiation	RadAvg	486	774	1118	1463	1705	1912	1930	1678	1366	895	567	397		
	RadStd	50	103	105	129	153	125	100	71	116	78	76	40		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page